

Revision on groups, rings & fields

$\Rightarrow \mathbb{R}^2 \cong \mathbb{C}$ and hence is a field.

fractal dimensions $\rightarrow$ eg a line that spans the plane but is not a plane $\therefore \underline{1 < d < 2}$.

info
• Show that every finite spanning set contains a basis.

$$\Rightarrow V = \text{span}(\{v_1, \dots, v_n\})$$

$$\text{if } v_j = \sum_{i \neq j} \alpha_i v_i$$

remove v_j from set

$\Rightarrow$ Repeat until only linearly independent elements are left.

$$\Rightarrow \{v_k, v_{k+1}, \dots, v_{k+r}\}$$

Th
• Show that every independent set can be extended to a basis.

$\Rightarrow \{v_1, v_2, \dots, v_k\}$ our set

$\{w_1, \dots, w_n\}$ a spanning set

$$\{v_1, \dots, v_k, w_1, \dots, w_n\}$$

$\hookrightarrow$ keep removing until all are independent

Matrices

$\mathbb{R}^{p \times q} \Rightarrow p \times q$ matrix.

$\Rightarrow$ matrix

data linear function

$$X \xrightarrow{A} Y$$

$Y = AX$

$\Rightarrow$ Basically a matrix also acts as a mapping function from one space to another space

The set of all real matrices with same dimension are a group.

Products of matrices $\rightarrow$

$\rightarrow$ Inner product

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$$

$$c_{ij} = a_i^T b_j$$

$\rightarrow$ Outer product

$$C = \sum a_k b_k^T$$

⇒ Row picture & column picture for matrix multiplication.

4

$\Rightarrow$ Read about the proof of associativity of matrix multiplication.

$$\# \quad A B \quad \Rightarrow \quad A \cdot B$$
$$\sim$$
$$(f(g(x)))$$

The order of multiplication of matrices would determine the number of multiplication ops required.

$A \rightarrow n \times m$
 $B \rightarrow m \times r$
 $C \Rightarrow r \times p$

$$\# \text{ ops } (AB)C = n \times r \times m + m \times p \times r$$

Linear Systems

⇒ System of linear equations

$$A x = b$$

$$A = \mathbb{R}^{n \times m}$$

$$x \rightarrow \mathbb{R}^m$$

$$b \rightarrow \mathbb{R}^n$$

⇒ Swapping rows is ok

⇒ multiply by non-zero

⇒ add a multiple of row to another.

Proof of last one

$$a_i^T x = b_i \quad // \quad \boxed{c a_j^T s'' = c b_j}$$

$$(a_i^T + c a_j^T) x = b_i + c b_j$$

Let says solutions are s & s'

$$\boxed{a_i^T s = b_i}$$

$$a_i^T s' + c a_j^T s' = b_i + c b_j$$

$$a_i^T (s' - s) + c a_j^T s' = c b_j$$

$$\underbrace{a_i^T (s' - s)}_{\neq 0} + c \underbrace{a_j^T (s' - s'')}_{\neq 0} = 0$$

$s = s'$
 $s' = s''$

//
 $\neq 0$

Gaussian elimination

Augmented matrix $\rightarrow \left[A \mid b \right]$

The aim is to get to identity on the right or upper triangular so that we can solve the system

$\Rightarrow$ go pivot by pivot -

$\rightarrow$ LU decomposition

$$LU = A$$

$\swarrow$ $\rightarrow$ upper Δ
 $\searrow$ lower Δ elimination matrix. (E^{-1})

Read about precision traps
in gaussian elimination or
other stuff affecting matrix
ops.

Rank

nullspace or kernel.

$$N(A) = \{x \mid Ax = 0\}$$

column space (image)

$$C(A) = \{Ax \mid x \in \mathbb{R}^n\}$$

$$\text{rank} = \dim(C(A))$$

$$\boxed{\text{nullity}(A) = \dim(N(A))}$$

$$\# \quad \text{rank}(A \cdot B) \leq \text{rank}(A)$$

↓

AB is linear combination of columns of A with weights in B .

$$\# \quad (A \cdot B)x = A(Bx)$$

↓

simpler proof as

↓, x

Bx can at most span the dimension of x

Revision

$$\begin{bmatrix} \varepsilon & 1 \\ 1 & 1 \end{bmatrix} x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$\Rightarrow$ Matlab spits out the correct
 $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$$\begin{bmatrix} 1 & 1 \\ \varepsilon & 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \varepsilon & 1 \\ 0 & 1 - \frac{1}{\varepsilon} \end{bmatrix} = \begin{bmatrix} 1 \\ 2 - \frac{1}{\varepsilon} \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 - \varepsilon \end{bmatrix} = \begin{bmatrix} 2 \\ 1 - 2\varepsilon \end{bmatrix}$$

$$\frac{\varepsilon - 1}{\varepsilon} \cdot x_2 = \frac{2 \cdot \varepsilon - 1}{\varepsilon}$$
$$x_2 = \frac{2 - \varepsilon - 1}{\varepsilon - 1}$$

Rank/Nullity

$$\underbrace{\text{nullity}(A)}_k + \text{rank}(A) = n$$

$\{u_1, u_2, \dots, u_k\} =$ Independent basis for $N(A)$

$$\{u_1, u_2, \dots, u_k, w_1, \dots, w_{n-k}\} \rightarrow A$$

$\hookrightarrow$ independent basis
for $\mathbb{R}^n$

$$\rightarrow \{Au_1, Au_2, \dots, Aw_1, \dots, Aw_{n-k}\}$$

$y \in \text{col}(A)$ then

$$y = Ax \quad \text{where}$$

$$x = \sum \alpha_i u_i + \sum \beta_j w_j$$

$$Ax = \sum \alpha_i \cancel{Au_i} + \sum \beta_j Au_j$$

$\nearrow 0$

Ax is spanned by
 $\{Aw_1, \dots, Aw_{n-k}\}$

→ Independence →

$$\sum \beta_i Aw_i = 0$$

$$\text{then } \sum \beta_i w_i \in \text{Null}(A)$$

but we had constructed the
set $\{u_1, \dots, u_k, w_1, \dots, w_{n-k}\}$
as a basis for $\mathbb{R}^n$

$$\sum \beta_i w_i = \sum \alpha_i u_i$$

So, $\beta_i = 0$ and hence the
set is independent.

$$\text{rank}(A) = \text{rank}(A^T)$$

$A \rightarrow k$ rank

$$A = CF \rightarrow \text{combination of}$$

ind. basis of $C(A)$ as cols.

Exercises

- $\text{Null}(A^T A) = \text{Null}(A)$

$$A^T A x = 0$$

$$A x = 0$$

$$\boxed{A^T A x = 0}$$

$$\boxed{N(A) \subseteq N(A^T A)}$$

Now to prove that if

$$A^T A x = 0 \quad \text{then} \quad Ax = 0$$

$$\Rightarrow x^T A^T A x = x^T 0$$

$$\Rightarrow (Ax)^T (Ax) = 0$$

$$\Rightarrow \boxed{Ax = 0}$$

$$N(A^T A) \subseteq N(A)$$

$$\text{Hence} \quad N(A^T A) = N(A)$$

- $A \in \mathbb{R}^{m \times n}$ & $B \in \mathbb{R}^{n \times p}$

$$A': \quad x \rightarrow Ax \quad \forall x \in B y$$

$$x \in \mathbb{R}^n \quad y \in \mathbb{R}^p$$

$$\text{in } A' : \quad x \rightarrow A(Bx)$$

$$\text{col}(A') = \text{col}(AB)$$

$$\text{ii) } \quad A(\underline{Bx}) = 0$$

$$Bx \in \text{Null}(A)$$

$$Bx \in \text{col}(B)$$

$$\Rightarrow \text{Null}(A) \cap \text{col}(B)$$

$$\text{iii) } \begin{array}{ccc} \text{rank}(A') + \text{nullity}(A') = \text{dim}(A') \\ \downarrow \quad \quad \downarrow \\ \text{rank}(AB) \quad \text{dim}(\text{Null}(A) \cap \text{col}(B)) \end{array} \quad \text{rank}(B)$$

$$\text{iv) } \text{dim}(\text{Null}(A) \cap \text{col}(B))$$

//

dimension of intersection would
be max if

$$\text{Null}(A) = \text{col}(B)$$

$$\text{Null}(A) = n - \text{rank}(A)$$

$$\rightarrow \boxed{\text{rank}(AB) \geq \text{rank}(A) + \text{rank}(B) - n}$$